CX 4220/CSE 6220 High Performance Computing
HW 1 Solutions

1.

$$S(p) = \frac{4n}{\frac{8n}{p} + 1000\log p}$$

$$E(p) = \frac{4n}{8n + 1000p\log p}$$

   (a) For $n >> p$, $S(p) \rightarrow \frac{4n}{\frac{8n}{p}} = \frac{p}{2}$.

   (b) $E(p) \geq \frac{1}{3} \quad \Rightarrow \quad 1000p\log p \leq 4n \quad \Rightarrow \quad p\log p \leq 10,000$
   $\Rightarrow \quad p^p \leq (2^{10})^{1000} \;\approx\; 1000^{1000}$
   $p_{\max} \;\approx\; 1000$

   (c) $n > 250 \times 64 \times \log 64 \;=\; 250 \times 64 \times 6 \;=\; 96,000.$

2. $E(p) = \Theta\left(\frac{n}{n\log n + p\log p}\right)$

   (a) Maximum efficiency possible is $\Theta\left(\frac{n}{n\log n}\right) = \Theta\left(\frac{1}{\log n}\right)$

   (b) For achieving maximum possible efficiency, $p\log p = \mathcal{O}(n\log n)$.
   Hence, $p \leq n \implies$ maximum of n processors can be used.

3. Assume exchange operation happens in O(1)

$$\text{Work} \;=\; \Theta(n^2)$$

`parallel for` can assign separate processor/thread for each iteration in the loop. This assignment of tasks under each parallel for loop will have span of $\Theta(\log n)$. Assuming that we have enough processors,

$$\text{Span} \;=\; \Theta(\log n) + \Theta(\log n) + O(1) \;=\; \Theta(\log n)$$

$$\text{Parallelism} \;=\; \frac{\text{work}}{\text{span}} \;=\; \Theta(\frac{n^2}{\log n})$$

4.

$$\text{Work} \;=\; \Theta(n^2)$$

`parallel for` can assign separate processor/thread for each iteration in the loop. This will have a span of $\Theta(\log n)$. The inner for loop will have a span of $\Theta(n)$. Assuming that we have enough processors,

$$\text{Span} \;=\; \Theta(\log n) + \Theta(n) \;=\; \Theta(n)$$

$$\text{Parallelism} \;=\; \frac{\text{work}}{\text{span}} \;=\; \Theta(n)$$

5. Parallel Floyd-Warshall Algorithm

Algorithm 1 Parallel Floyd-Warshall (W,n)

$D^{(0)} = W$
for k from 1 to n
    let$D^{(k)} = (d_{ij}^{(k)})$be a new $n \times n$ matrix
    parallel for i from 1 to n
        parallel for j from 1 to n
            $d_{ij}^{(k)} \;=\; min\{d_{ij}^{(k-1)},\; d_{ik}^{(k-1)} + d_{kj}^{(k-1)}\}$
return $D^{(n)}$

As value of $d_{ij}^{(k)}$ is dependent on the matrix values at $(k-1)$, we cannot parallelize the outermost k loop.

Work $= \Theta(n^3)$
Span $= \Theta(n \log n)$
Parallelism $= \frac{n^2}{\log n}$